

**THE SECOND WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION
March 4, 1939**

Morning Session

1. Find the length of the curve $y^2 = x^3$ from the origin to the point where the tangent makes an angle of 45° with the x -axis. (page 95)

2. A point P is taken on the curve $y = x^3$. The tangent at P meets the curve again at Q . Prove that the slope of the curve at Q is *four* times the slope at P . (page 95)

3. Find the cubic equation whose roots are the cubes of the roots of
(page 96)

$$x^3 + ax^2 + bx + c = 0.$$

4. Find the equations of the *two* straight lines each of which cuts all the *four* straight lines

$$x = 1, y = 0; \quad y = 1, z = 0; \quad z = 1, x = 0; \quad x = y = -6z.$$

(page 98)

5. Take either (i) or (ii).

(i) Solve the system of differential equations

$$\frac{dx}{dt} = x + y - 3,$$

$$\frac{dy}{dt} = -2x + 3y + 1,$$

subject to the conditions $x = y = 0$ for $t = 0$. (page 101)

(ii) A heavy particle is attached to the end A of a light rod AB of length a . The rod is hinged at B so that it can turn freely in a vertical plane. The rod is balanced in the vertical position above the hinge and then slightly disturbed. Prove that the time taken to pass from the horizontal position to the lowest position is

$$\sqrt{\frac{a}{g}} \log_e (1 + \sqrt{2}).$$

(page 103)

6. Take either (i) or (ii).

(i) A circle of radius a rolls on the inner side of the circumference of a circle of radius $3a$. Find the area contained within the closed curve generated by a point on the circumference of the rolling circle. (page 105)

(ii) A shell strikes an airplane flying at a height h above the ground. It is known that the shell was fired from a gun on the ground with a muzzle velocity of magnitude V , but the position of the gun and its angle of elevation are both unknown. Deduce that the gun is situated within a circle whose center lies directly below the airplane and whose radius is

$$\frac{V}{g} \sqrt{V^2 - 2gh}.$$

(Neglect the resistance of the atmosphere.)

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7. Take either (i) or (ii).

(i) Find the curve touched by all the curves of the family

$$(y - k^2)^2 = x^2(k^2 - x^2).$$

Make a rough sketch showing this curve and two curves of the family.

(page 109)

(ii) If the expansion in powers of x of the function

$$\frac{1}{(1 - ax)(1 - bx)}$$

is given by

$$c_0 + c_1x + c_2x^2 + c_3x^3 + \dots,$$

prove that the expansion in powers of x of the function

$$\frac{1 + abx}{(1 - abx)(1 - a^2x)(1 - b^2x)}$$

is given by

$$c_0^2 + c_1^2x + c_2^2x^2 + c_3^2x^3 + \dots.$$

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Afternoon Session

8. From the vertex $(0, c)$ of the catenary

$$y = c \cosh \frac{x}{c}$$

a line L is drawn perpendicular to the tangent to the catenary at a point P . Prove that the length of L intercepted by the axes is equal to the ordinate y of the point P . (page 114)

9. Evaluate the definite integrals

$$(i) \int_1^3 \frac{dx}{\sqrt{(x-1)(3-x)}}, \quad (ii) \int_1^\infty \frac{dx}{e^{x+1} + e^{3-x}}.$$

(page 114)

10. Given the power-series

$$a_0 + a_1x + a_2x^2 + \dots$$

in which

$$a_n = (n^2 + 1)3^n,$$

show that there is a relation of the form

$$a_n + pa_{n+1} + qa_{n+2} + ra_{n+3} = 0,$$

in which p, q, r are constants independent of n . Find these constants and the sum of the power-series. (page 115)

11. Find the equation of the parabola which touches the x -axis at the point $(1, 0)$ and the y -axis at the point $(0, 2)$. Find the equation of the axis of the parabola and the coordinates of its vertex. (page 117)

12. Take either (i) or (ii).

(i) Prove that

$$\int_1^a [x]f'(x) dx = [a]f(a) - \{f(1) + \dots + f([a])\},$$

where a is greater than 1 and where $[x]$ denotes the greatest of the integers not exceeding x . Obtain a corresponding expression for

$$\int_1^a [x^2]f'(x) dx. \quad (\text{page 121})$$

(ii) A particle moves on a straight line, the only force acting on it being a resistance proportional to the velocity. If it started with a velocity of 1,000 ft. per sec. and had a velocity of 900 ft. per sec. when it had travelled 1,200 ft., calculate to the nearest hundredth of a second the time it took to travel this distance. (page 122)

13. Take either (i) or (ii).

(i) Let $f(x)$ be defined for $a \leq x \leq b$. Assuming appropriate properties of continuity and derivability, prove for $a < x < b$ that

$$\frac{\frac{f(x) - f(a)}{x - a} - \frac{f(b) - f(a)}{b - a}}{x - b} = \frac{1}{2} f''(\xi),$$

where ξ is some number between a and b .

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(ii) Calculate the mutual gravitational attraction of two uniform rods, each of mass m and length $2a$, placed parallel to one another and perpendicular to the line joining their centers at a distance b apart.

In your answer let a approach zero, and comment on the form of the result.

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14. Take either (i) or (ii).

(i) If

$$u = 1 + \frac{x^3}{3!} + \frac{x^6}{6!} + \dots,$$

$$v = \frac{x}{1!} + \frac{x^4}{4!} + \frac{x^7}{7!} + \dots,$$

$$w = \frac{x^2}{2!} + \frac{x^5}{5!} + \frac{x^8}{8!} + \dots,$$

prove that

$$u^3 + v^3 + w^3 - 3uvw = 1.$$

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(ii) Consider the central conics

$$(ax^2 + by^2) + 2(px + qy) + c = 0,$$

$$(ax^2 + by^2) + 2\lambda(px + qy) + \lambda^2 c = 0,$$

where λ is a given positive constant.

Show that if all radii from the origin to the first conic are changed in the ratio λ to 1 the tips of these new radii generate the second conic.

Let P be the point with coordinates

$$x = -\frac{p}{a} \frac{2\lambda}{1 + \lambda}, \quad y = -\frac{q}{b} \frac{2\lambda}{1 + \lambda}.$$

Show that if all radii from P to the first conic are changed in the ratio λ to 1 and then reversed about P the tips of these new radii generate the second conic.

Comment on these results in case $\lambda = 1$.

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